KENNETH SAUERS

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EXPERIENCE

Software Engineer, System Innovation Group **Dec 2020 – Current**

- Developed a high-accuracy machine learning classifier (98%) to detect base stations, strengthening network security.
- Engineered an AM transmission system using SDR, enabling controlled web-based communication over 200 miles.
- Spearheaded the consolidation of multiple applications into a unified framework, streamlining deployment and accelerating prototyping.
- Authored and released comprehensive technical documentation including Interface Control Documents (ICDs), design specifications, and tooling guides for internal teams and external clients to ensure seamless integration and support.

Software Engineer, Department of Defense - NSIN **May 2020 - Sep. 2020**

- Collaborated one-on-one with active-duty airmen to develop a simulation tool for KC-135 aircraft maintenance, enhancing training realism.
- Designed and prototyped a VR application to train aircraft maintainers, improving hands-on learning experiences.
- Processed and optimized KC-135 point cloud data for VR simulations, ensuring accurate and high-fidelity aircraft modeling.

Research Assistant, Center for computer vision research lab UCF **Sep. 2017 - Mar. 2020**

- Developed a synthetic data generation tool for machine learning, increasing testing accuracy by 13%.
- Contributed to research on Generative Adversarial Networks (GANs) and their applications in Computer Vision, enhancing data augmentation and model performance.

EDUCATION

Master of Science in Computer Science, Georgia Institute of Technology **2025**

- Specialization in Computational Perception and Robotics

Bachelor of Science in Computer Engineering, University of Central Florida **2021**

- Minor in Intelligent Robotic Systems
- Robotics Team, Computer Vision Engineer
- Competitive Programming Team, Programmer

PROJECTS

Personal Project | BDLC Actuator **2024**

Developed a robotic actuator with Field-Oriented Control (FOC) for precise position, velocity, and torque regulation via a CAN interface.

- Designed a custom PCB for motor control and communication
- Implemented daisy-chainable CAN communication ports for modular integration
- Engineered a 46:1 gear reduction using a custom 3-stage compound planetary gearbox

Swamphacks | Skin Disease Diagnosing Mobile App **2020**

A computer vision app that detects skin disease via a convolutional neural network

- **Award**, Most Technically Impressive
- **Technology**, Convolutional Neural Network, OpenCV, Pillow, Native Android, Flask, and GCP

SKILLS

- **Certifications** Coursera: Deep Learning Specialization, CIW: JavaScript Specialist
- **Languages** C#, Python, TypeScript, JavaScript, Java, C++
- **Frameworks** .Net, ASP.NET, Entity Framework, React, Blazor, FastAPI, ROS, OpenCV
- **Technology** Docker, DevOps, CI/CD, SQL, GnuRadio
- **Strengths** Web Applications, Hardware Integration, Reverse Engineering, Machine Learning, Computer Vision